



AKGEC SKILLS FOUNDATION

Training Program Details

CENTRE NAME - NI LabVIEW Academy

Training/ Course Title:	Winter Training Program on Industrial IoT with LabVIEW
Target Audience:	Students
Fee:	7000/-
Duration:	40 Hrs.
Module	• Introduction to Industrial IoT concepts, LabVIEW basics, and the dataflow programming model. • Core LabVIEW programming: controls/indicators, loops, case structures, shift registers, and basic debugging. • Data acquisition fundamentals with NI DAQ: analog/digital I/O, sampling, signal conditioning, and simple measurements. • Hands-on myRIO applications: onboard I/O, real-time acquisition, PWM, and closed-loop control basics. • File I/O, logging and visualization: CSV/TDMS logging, charts, waveform processing, and basic analysis. • Python & LabVIEW integration plus IoT protocols: calling Python from LabVIEW, MQTT/HTTP basics, and cloud data exchange. • Edge computing, security considerations, mini hardware project, and final demonstration with documentation. • Basics of circuit schematic creation, PCB layout design, routing, and preparing files for fabrication.
Outcome:	• Learn to develop Industrial IoT applications using NI LabVIEW and embedded hardware. • Perform real-time data acquisition, logging, and monitoring for industrial scenarios. • Interface sensors and actuators using NI DAQ, myRIO and Arduino. • Build modular measurement and control applications with LabVIEW structures. • Analyze acquired data using charts, graphs, and waveform tools for decision-making. • Integrate Python with LabVIEW for automation, computation, and smart data processing. • Use IoT communication protocols such as MQTT/HTTP for sending and receiving data. • Implement basic edge computing workflows for local processing and reduced cloud load. • Troubleshoot hardware–software integration issues using systematic debugging. • Develop and present a mini IoT project demonstrating measurement, processing, and communication skills. • Gain foundational skills in PCB schematic design, layout creation, and preparing fabrication-ready PCB files.

Certification:	The participants, on successful completion of the above training, will receive certification by the NI LabVIEW Academy	
Program Coordinator:	Name: Ms. Deepti Singh	Contact number: 9818212869
Payment Mode:		